

DSCI-428 Group Project

Portfolio Optimization & Forecasting Project

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Introduction

Effective investment management increasingly depends on the ability to translate complex financial data into structured, actionable guidance.

Prescriptive analytics provides a rigorous framework for this process by not only evaluating historical performance and current conditions while also recommending specific decisions that optimize desired outcomes. This report applies prescriptive analytics techniques to construct and evaluate portfolio optimization models designed to support disciplined, data-driven investment allocation.

The central objective of our analysis is to determine the proportion of a client's funds that should be allocated to each stock in order to achieve an expected portfolio return at least equal to the average of the stocks' historical average annual returns, while simultaneously minimizing the overall variance of the portfolio's total return.

In addition, the report provides a return forecast for META as our selected stock; to illustrate how forward-looking insights can further inform prescriptive

recommendations. The overarching goals are to achieve a portfolio expected return greater than or equal to the average of the individual stock averages, minimize portfolio variance, and maintain meaningful diversification by limiting any single position to a maximum of 25 percent and ensuring at least seven active investments. Together, these analyses demonstrate how prescriptive analytics can guide investment decisions with precision, transparency, and strategic discipline.

Data Methodology

The portfolio optimization analysis relied on a structured set of financial inputs designed to quantify both expected return and risk. We imported historical monthly returns from our selected stocks over the preceding 5 years, and we then used this data to calculate annual returns. These returns form the basis for computing the covariance matrix, which captures the joint variability among the assets and provides the risk-interaction structure needed for mean-variance optimization. Collectively, these components establish the statistical foundation for formulating and solving the allocation problem.

Historical price data was obtained from the Bloomberg Terminal, a platform recognized for delivering high-quality and comprehensive financial information. To ensure data reliability, consistency was verified to make sure each month was accounted for within a five-year timeline of data we pulled.

The methodological approach centers on applying prescriptive analytics to construct optimization models aligned with clearly defined investment objectives. The covariance matrix was then generated to capture the variance and co-variance structure across assets. Using these inputs, a series of mean-variance optimization models were formulated. The base model seeks to achieve an expected portfolio return at or above the average of the stocks' historical annual returns while minimizing total portfolio variance. Additional models incorporate diversification constraints by limiting individual positions to no more than 25 percent of the portfolio and requiring a minimum of seven active investments and alternative risk-tolerance specifications to generate both risk-adverse and moderate-risk allocation strategies.

All models were solved using standard mathematical programming techniques, enabling the evaluation of feasible allocations that satisfy the stated return, risk, and diversification criteria. This methodological framework ensures

a transparent, replicable, and analytically rigorous process for deriving prescriptive investment recommendations.

Model Development

As stated previously, we seek to develop multiple different portfolios optimized to meet the diverse needs of our various clients. Our base model (front-end setup) sought to allocate the customer's funds to achieve a return greater than or equal to the average annual returns for each stock, while minimizing the portfolio's total return variance. Other constraints were also built into the model, such as each individual stock in the portfolio cannot be less than zero (to bar the model from short selling). We also ensured that the entirety of the portfolio was invested, so as to ensure none of the client's money was left idle. This model achieves returns of 20.47% per annum with a standard deviation (square root of variance) of 3.2%.

Equations

Objective Function: $\text{MIN} = \text{VAR} (\sum \text{each individual stock} * \text{interaction of variances})$

Decision Variables: Stock meta + Stock xom Stock hd

Constraints: ER (expected return) $\geq$ RR (required return)

Each individual stock ≥ 0

$$\Sigma(\text{stock meta} + \text{stock xom} + \dots) = 1$$

Constrained Version of Base Model

The decision variables and objective function for this version are unchanged from the previous version. What has changed is the addition of more constraints, hence its name. The added constraints are a requirement that at least 7 different companies be represented in the model. To make this happen in Excel, having an investment in a company is converted to a binary variable. We also ensured that no individual company could make up more than $\frac{1}{4}$ of the portfolio. The portfolio achieves comparable returns to the base model.

Equations for additional constraints

$$\text{Stock meta} + \text{Stock xom} + \text{Stock nvda} \dots \geq 7$$

$$\text{Stock meta} \leq 0.25, \text{Stock xom} \leq 0.25, \text{Stock nvda} \leq 0.25, \text{etc.}$$

Risk Aversion Models

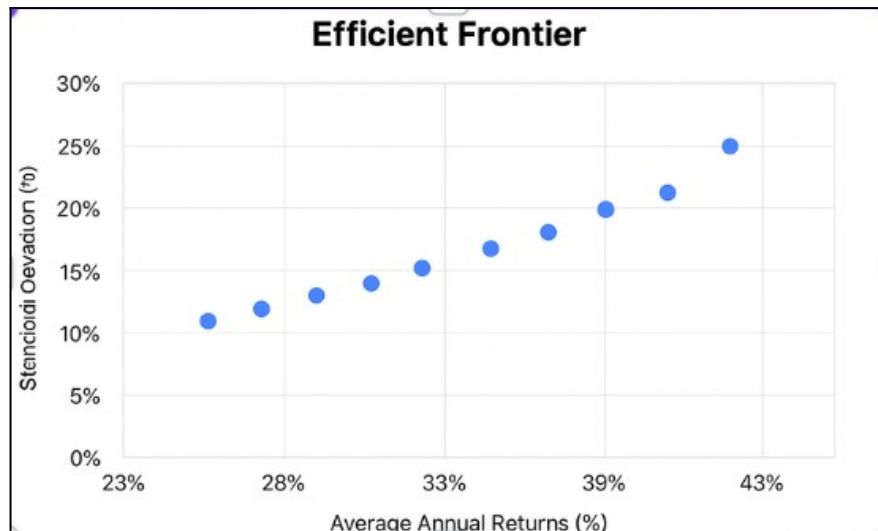
These models differ from the first two models in that the objective function they work towards optimizing is a weighted measure that takes both expected return and the client's aversion of risk into account. Thus, we seek to maximize this

measure. Risk aversion is fed into the model using a factor of r , which the client determines.

Markowitz mean-variance model

The mean-variance model employed here was originally developed by Harry Markowitz in the 1950s. It describes the efficient frontier of securities (the combination that produces the best returns for any given risk tolerance). The version of this that we employ, $E(R) - r\sigma^2$, is consistent with the original formulation of risk aversion. Our version only differs from the class textbook's reformulation in that our r is not bounded between 0 and 1 but instead can take any positive number (higher being more averse to risk). This does not change the economic interpretation of the model, and feeding different values into the model can be used to sketch out a graph of the efficient frontier. We then convert variance to standard deviation for easier interpretation.

Figure 1. Our Portfolio's Efficient Frontier



Moderate Risk Tolerance

For the moderate risk portfolio, we selected an r of 2. This portfolio saw volatility fall from 25.2% to 19.8%, whereas returns fell from 41% to 39% over the 0.5 baseline. Thus, this client is losing a tiny bit of return in exchange for a big reduction in his risk, making this a reasonable compromise.

Greater Risk Aversion

For the greater risk aversion portfolio, we selected an r of 8. This portfolio sees volatility fall to 8.8%, and annual returns fall to 26.6% a year. Thus, most of the risk has been eliminated compared to the moderate portfolio, and yet the client is still able to achieve healthy returns.

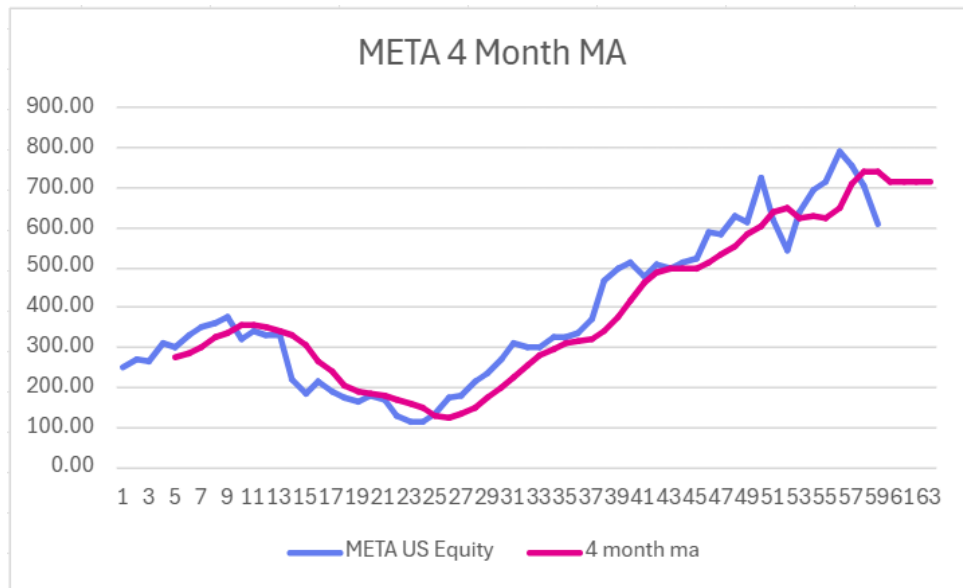
If one is curious about why these risk aversion portfolios seem to have more risk compared to the baseline model, it is because they are designed to maximize

returns while keeping risk aversion into account. Whereas the baseline model only seeks to minimize variance. If we were to compare these risk averse portfolios to a hypothetical portfolio composed of these stocks that maximize expected returns, we would easily observe risk reduction taking place.

Results & Recommendations

The results show distinct variations in performance among the three forecasting techniques used for META's monthly stock prices. With an MSE of 3,935.37, the simple four-month moving average yielded the worst results, demonstrating its inability to react quickly to the unpredictability and rising trend shown in META's recent price movements. This approach reacts to changes in the data too slowly, resulting in more forecast errors, as it gives equal weight to each of the four most recent observations.

Figure 2. META Stock Price vs. 4-Month Moving Average



With an MSE of 2,005.08, the weighted four-month moving average did slightly better. The forecasts' accuracy increased when more recent months were given more weight, but the model's ability to adjust to shifting trends was still constrained by the set weighting structure. Therefore, this approach was not as responsive as needed to identify a stock displaying significant trend activity, but it was still more successful in capturing the direction of the data than the simple moving average.

With the lowest MSE of 1,950.80, the double exponential smoothing (Holt) approach was the most accurate model. This method allows the model to more easily adapt to META's recent rising trend by incorporating both a level and a trend component. Holt's approach is the most dependable method for this dataset, roughly it offers the best balance between catching the underlying trend and reducing short-term fluctuations, as seen by the decreased predicting error. Based

on this model, the expected stock values for December 25, January 26, February 26, and March 26, are all 714.98. Because the trend and level components stabilized near the end of the sample, the prediction remained constant over the four-month period.

Considering these results, the double exponential smoothing model is recommended as the most suited forecasting approach for META. Compared to both moving average approaches, which were unable to keep up with the recent price movements, it is the most suitable option due to its capacity to adapt to changes in trends. The estimated value of about \$714.98 indicates that META could stay at its present level through early 2026. This projection indicates that META can be a fair component of a diverse mix of equities for the customer's portfolio construction. However, the forecast should not be considered on its own. It must be considered when paired with the outcomes of the optimization model, which considers the yearly returns, risk levels, and collective movements of each stock. When all of these factors are taken into consideration, the company may select a portfolio that meets the objectives of the client while still achieving great performance and risk management. The trade-off between predicted yearly returns and portfolio risk is depicted by Figure 1 on page 7, which shows the efficient frontier produced by the optimization model. The graph confirms that the

recommended portfolio distribution lies along the efficient frontier and gives the greatest anticipated return for its level of risk.

Conclusion

This study demonstrated how integrating optimization modeling, risk analysis, and forecasting tools may help clients with various goals to make data-driven investment decisions. Using five years of stock data, we developed basic, restricted, and risk-averse optimization models to balance individual risk preferences, diversification requirements, and expected returns. Our forecasting study, which discovered that Holt's double exponential smoothing provided the best accurate prediction of META's future price, further supported our recommendations. When compared to the efficient frontier, the selected strategies were found to offer the best returns for their respective risk levels. Overall, we were able to develop investment strategies that are both analytically solid and in line with our customers' long-term financial objectives by combining optimization, forecasting, and risk management approaches.